Develop AI Applications for Code Generation

Estimated time needed: 20 minutes

Welcome to the hands-on lab: Develop AI applications for code generation!

You learned that the foundation models can serve as a base to develop applications for text-related tasks like text completion, translation, and summarization. In this lab, you will leverage foundation models' capabilities of foundation models that can help you develop an app based on the code generated.

For this, you must determine the generative AI foundation model that best suits your use case. This will help you make informed decisions about the direction of your AI application development.

In this lab, you will generate code using the foundation model **GPT-5 Nano** to develop an AI application.

Learning Objectives

- Generate code for the desired programming language using GPT-5 Nano foundation model to develop simple AI-powered programming apps

How can you develop AI applications with generative AI foundation models?

Exploring various foundation models serves as a starting point for various AI applications. Understanding and examining their capabilities will help you draw a roadmap for developing AI applications.

Using an appropriate foundation model ensures you choose the right tools and technologies, saving time, effort, and resources.

In this lab, you will explore foundation models by finding answers to specific questions, such as:

- How can you develop an AI app that accomplishes a specific task with a specific code language?

    You can leverage the code generation capability of generative AI foundation models to write code in the C language that sorts elements of an array in ascending order or Python code that identifies whether a given number is prime.
    Let's get started prompting and experimenting with foundation models.

About generative AI classroom lab

▶ Click here

Notes:

1. The prompts used in this lab are for your reference only. You can create your own prompts and generate responses using generative AI.
2. Since AI-generated outputs are dynamic, you may receive different responses even though you've used the same prompt from this lab.

Before you begin prompting

Before beginning the exercise, you must set up the AI classroom and know the Generative AI workspace.

1. As a first step, you must set up your AI classroom for better learning experience.

- Name the chat
- Choose the foundation model

2. Know your workspace to prompt the foundation model-based chatbot better.

- Use prompt instructions
- Type your message
- Manage your chat
- Delete the chat from history

Exercise: Generate code using the foundation model to develop a simple AI app

Let's explore the code capabilities of foundation models that can help you develop an app based on the code generated.

In this exercise, you will generate code using the foundation model **GPT-5 Nano** to develop an AI application.

Step 1: Set your AI environment and select the foundation model

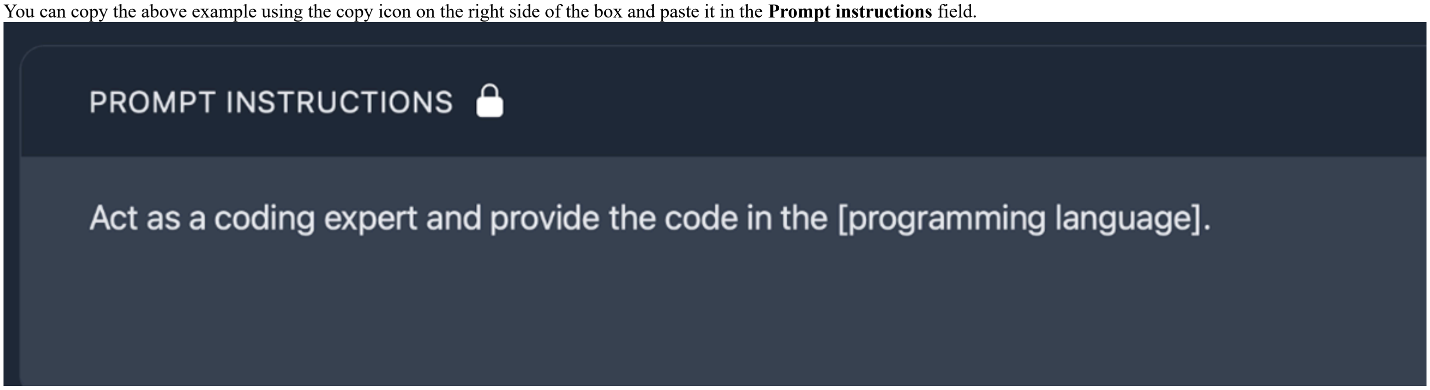
1. Create a new chat
2. Name the chat per the context
3. Select the foundation model **GPT-5 Nano**

Step 2: Write a relevant prompt for code generation

1. Consider the text you need to translate.
2. Provide the instructions in the **Prompt instructions** field.
   For example,

```
Act as a coding expert and provide the code in the [programming language].
```

You can copy the above example using the copy icon on the right side of the box and paste it in the **Prompt instructions** field.

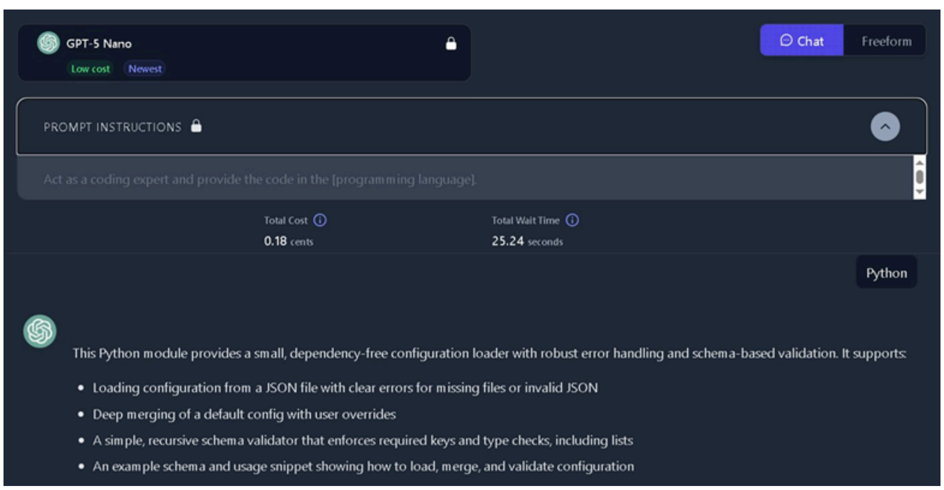


Note: You can include more instructions central to your context in the **Prompt instructions** field.

3. Write the programming language as a prompt in **Type your message** field.
   For example,

   ```
   Python
   ```

4. Click **Start Chat** and observe the response generated, as shown below.

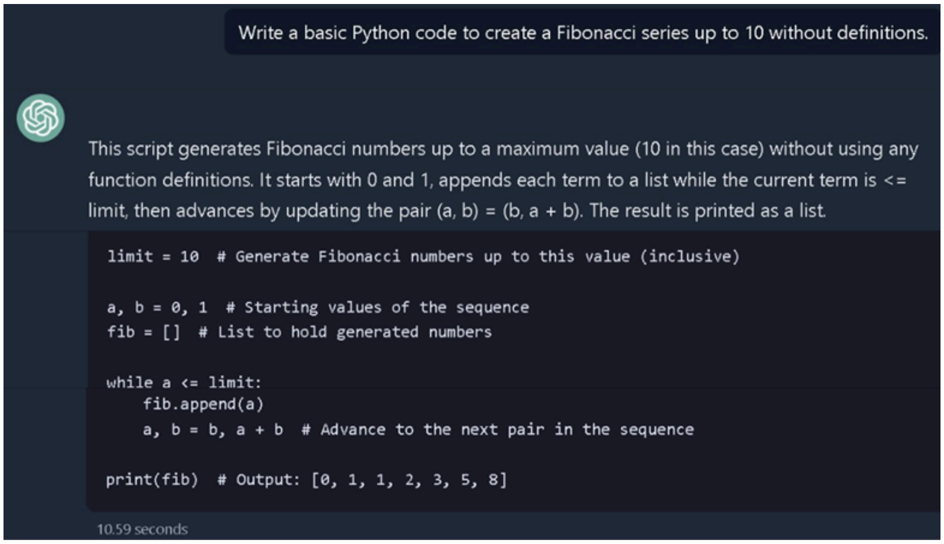


5. Now provide the prompt to generate the code for the desired context.
   For example,

   ```
   Write complete Python code to create a Fibonacci series up to 10.
   ```

   Note: The versatility of Fibonacci numbers and the Fibonacci sequence makes them a fascinating topic in mathematics and its applications across various fields. You can use the Fibonacci series to develop AI applications related to coding education, automated code generation, algorithmic problem-solving, and many more.

6. Click **Start Chat** and observe the response generated, as shown below.



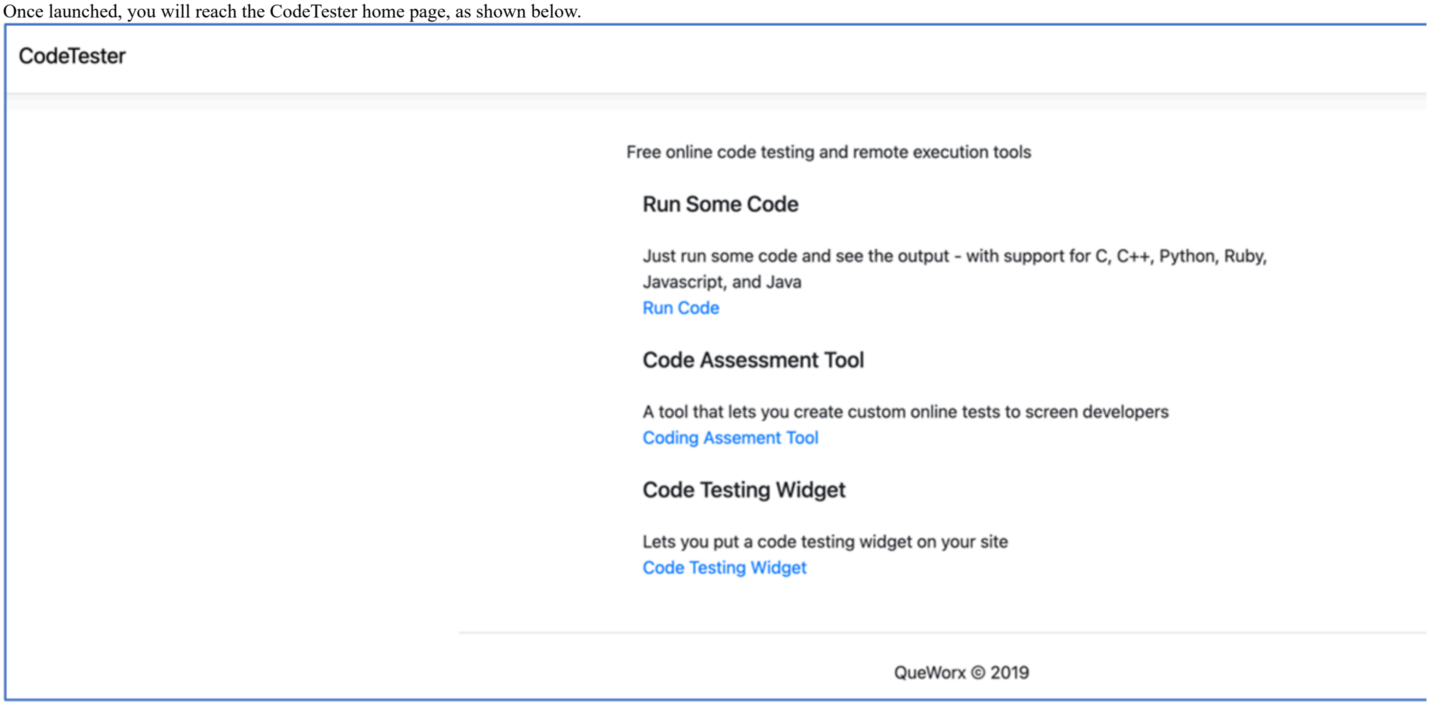
Note: While using the generated code, you must ignore the first line “python” and the last line “” of the code.

Step 3: Test the generated code

Testing the generated code helps you validate the accuracy and reliability of the code. You can use any accessible online programming language compiler tool to test the code. Here, CodeTester has been used to test and validate the generated code.

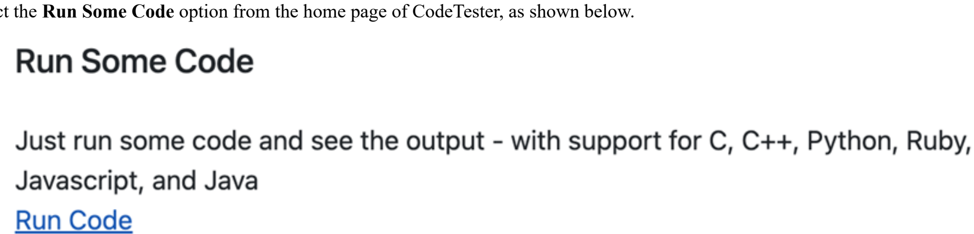
CodeTester is a free online code testing tool that can help you run the code, assess it, and execute it on your site.

1. Click CodeTester to launch it.
2. Once launched, you will reach the CodeTester home page, as shown below.



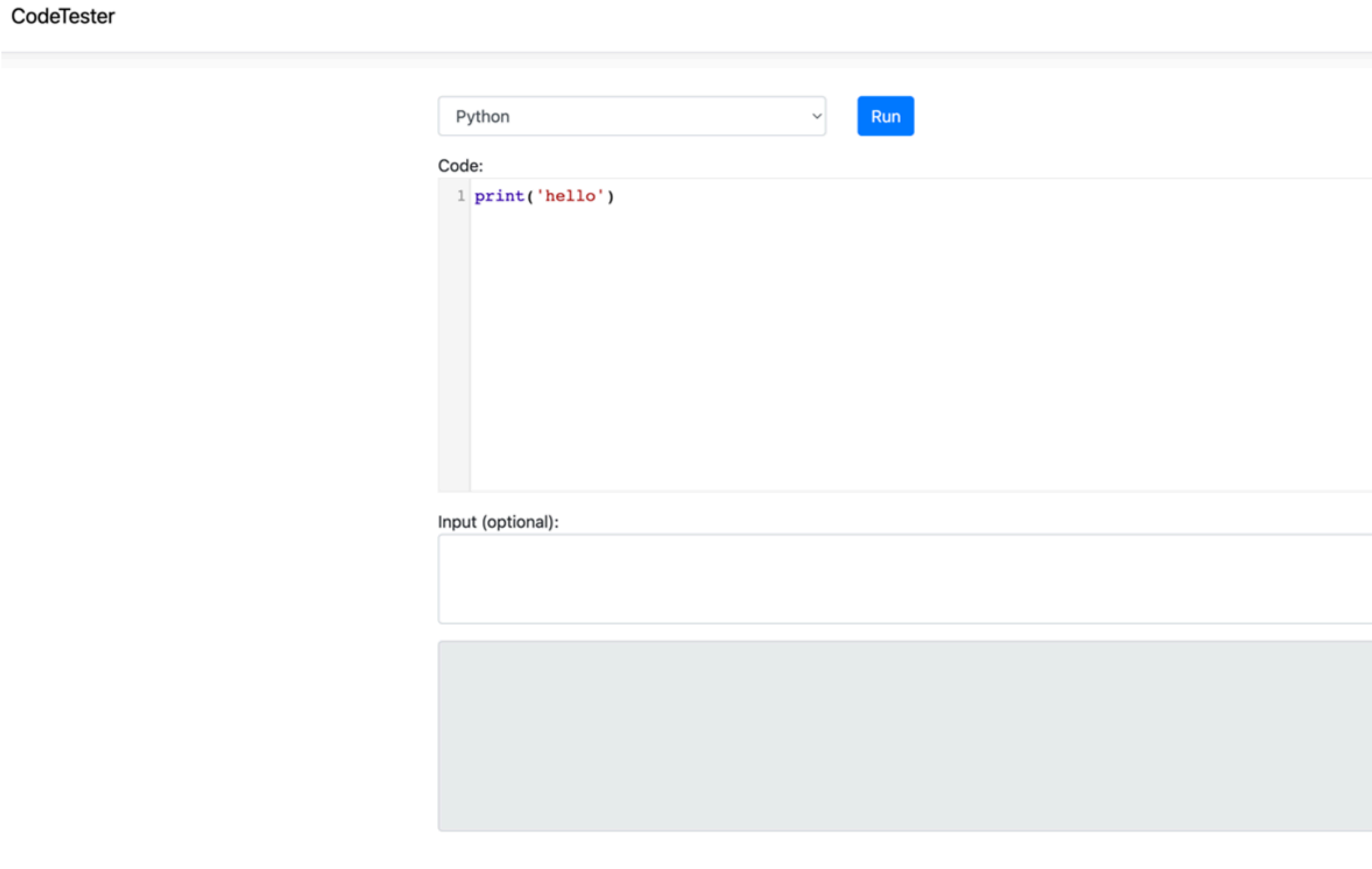
3. To reach the code testing page, you must follow any of the steps below.

- i. Select the **Run Some Code** option from the home page of CodeTester, as shown below.

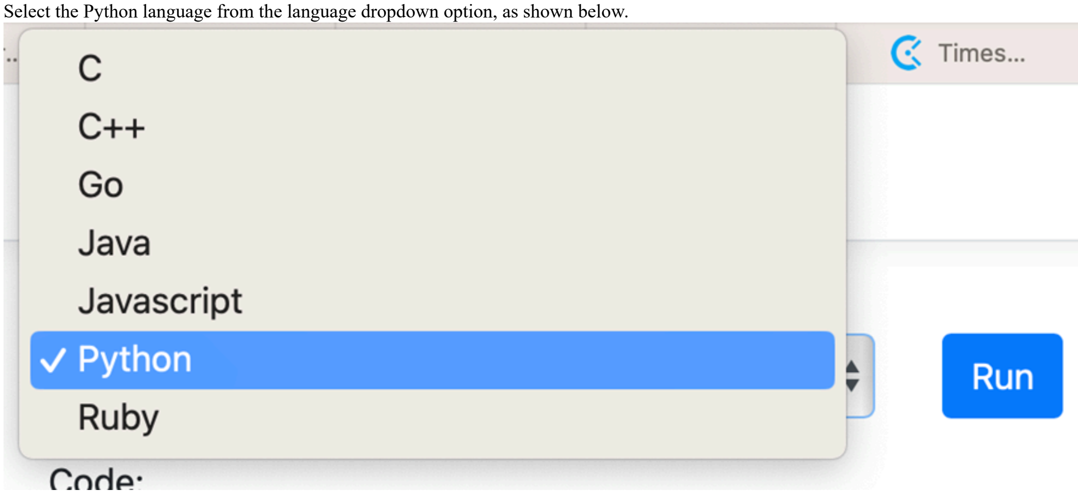


- ii. Alternatively, click CodeTester runner page.

4. On the CodeTester testing page, you can see the test pane, the optional input box, and the grey color output box, as shown below.



5. Select the Python language from the language dropdown option, as shown below.



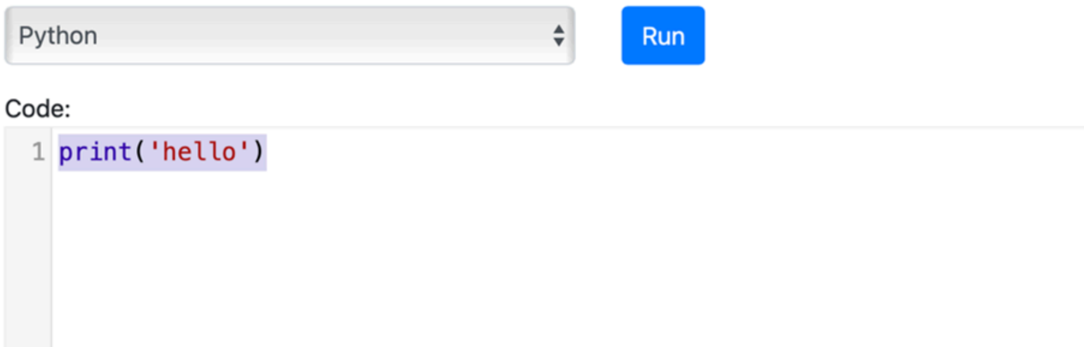
Note: You can choose the code language for which you have generated the code.

6. Now, go to the lab and copy only the code characters from the generated response in Step 2.
For example,

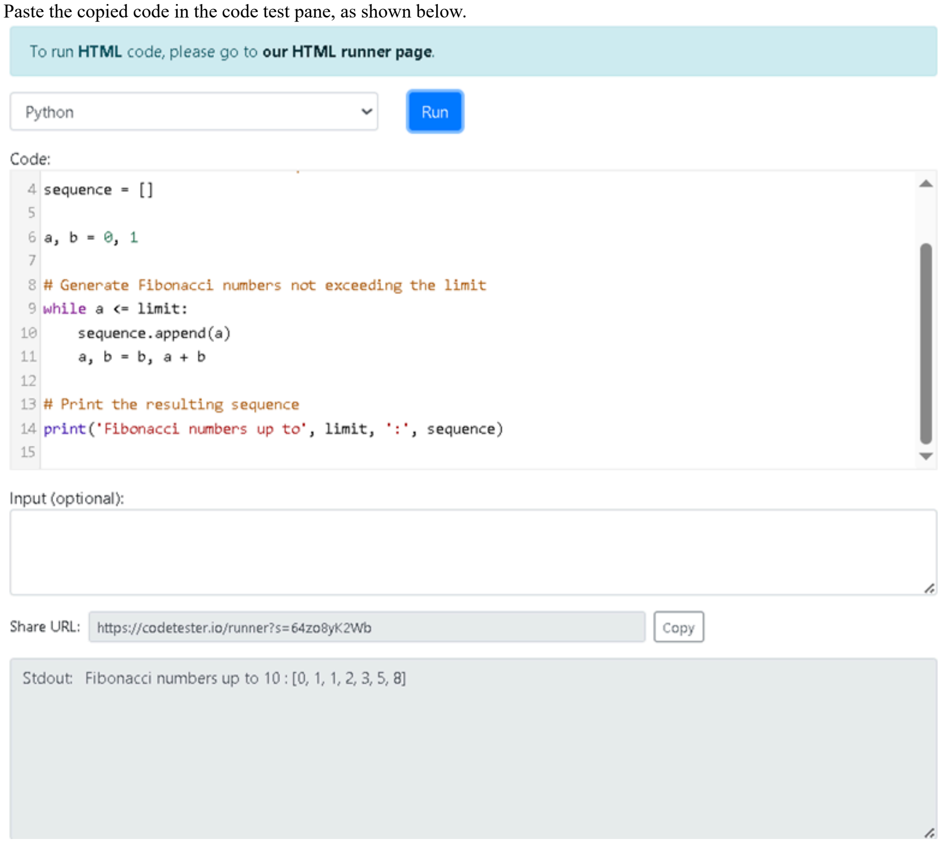
```
limit = 10
 # Initialize the Fibonacci sequence with the first two numbers
sequence = []
 a, b = 0, 1
 # Generate Fibonacci numbers not exceeding the limit
while a <= limit:
    sequence.append(a)
    a, b = b, a + b
 # Print the resulting sequence
print('Fibonacci numbers up to', limit, ':', sequence)
```

You can copy the above code using the copy icon on the right side of the box.

7. Go to the CodeTester testing page and delete the default code as highlighted in the screenshot below.

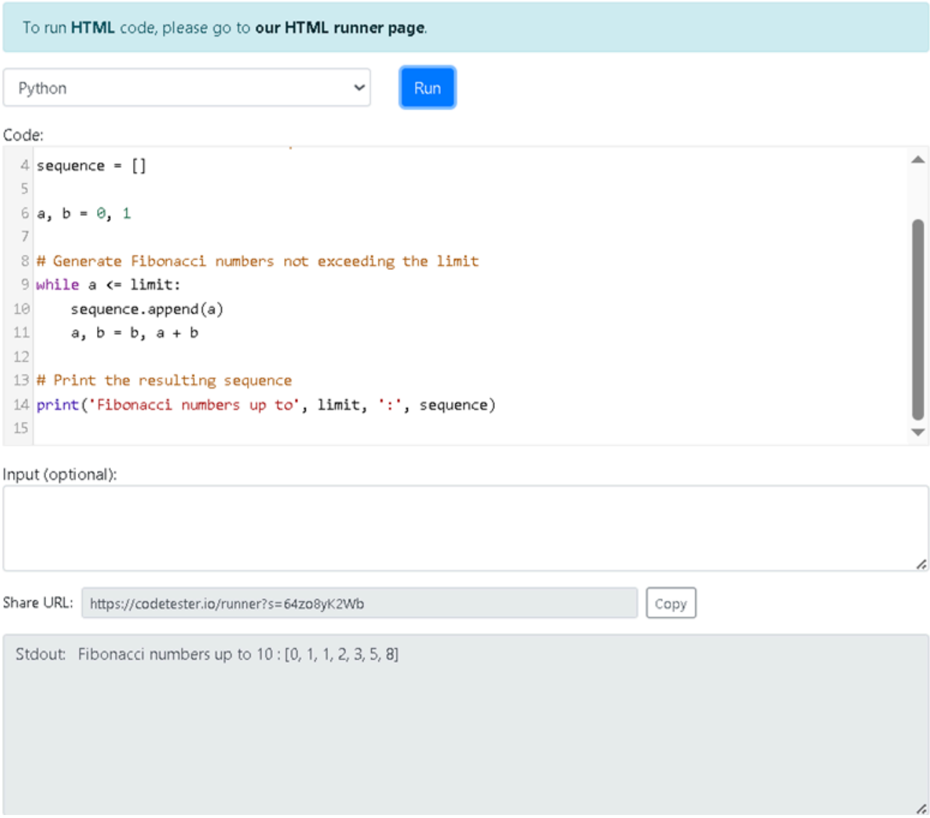


8. Paste the copied code in the code test pane, as shown below.



9. Click **Run** next to the language dropdown at the top to test and validate the code.

10. Review the results displayed in the lower box on the screen, as shown below.



Step 4: Review the scope and applications

You can use the GPT-5 Nano foundation model interface to generate more and more code for your programming requirements.

> Note: You must validate the code generated through ChatGPT for factual accuracy and use the technology ethically and responsibly.You can also learn coding and programming as the model provides step-by-step explanations and guidance. However, you cannot generate large and complex code from scratch, as their training data set based on 2021 libraries may limit their capability.

Summary

Congratulations! You just completed the hands-on lab to develop AI applications for code generation.

In this lab, you experimented with the code generation capability of the foundation models to explore its use in developing an AI app.

To leverage GPT-5 Nano for these purposes, you must use the OpenAI API and integrate it into your language learning or AI app development project. The API allows you to interact with the model and access its language generation capabilities programmatically, making it easier to create tailored language learning experiences or develop AI-powered language-related applications.